Forward Deployed Engineer assignment · Kinyanjui Njoroge · August 2026 · Live bulletin: sand-fde-bulletin.pages.dev · Repo: github.com/nyashkn/sand-fde-int · Full working notes, runbook, data contract and decision register are in the repository.

> *All figures below are computed on **synthetic data** — the assignment CSVs plus a hand-built DHIS2 sample — not real Rwanda health records. Facility mortality of 14.3–73.2 per 1,000 live births runs three to four times real Rwandan rates.*

The choice: Problem A, and why not B or C

Option	Verdict	Why
A — Quarterly Health Bulletin, automated	**Chosen**	Only option finishable and provable in 6 weeks; operates on data that already arrives, in a shape that already exists. The trust wedge for later work.
B — Real-time facility dashboard	Out of reach this sprint	Needs facility-level freshness. Reporting substrate is monthly aggregates on a 2–3wk lag — a granularity/cadence gap, not a coverage gap, regardless of how many facilities report.
C — Unified patient view (TB/HIV)	Rejected	Needs a clinical-safety hazard analysis that does not fit in 6 weeks. Propose a weekly reconciliation report instead — flags likely co-infection for human review, no record merging, none of the risk.

How this survived review. The choice was prompted hostilely against three independent model families (GPT-5.6-sol, Gemini 3.1 Pro, Grok 4.5). All three independently chose Problem A — but rejected the original justification, unanimously finding errors (an impossible 5-day publication target given the 2–3wk upstream lag; an overstated ROI baseline; a false claim that A's mart is "most of" B's data layer). All corrected in the full document.

What was built

A pipeline that reads the five provided CSVs plus a DHIS2-shaped sample, resolves every field through a 71-row declared crosswalk, and publishes all four 2024 quarters as self-contained HTML with zero client-side JavaScript. Python + Hamilton + DuckDB + Parquet (bronze → silver → gold → checks); Astro + Vega-Lite compiled to static SVG for render. 5 automated checks gate publication, each one added because it caught a real defect.

Two findings that shaped it more than any metric in the brief

1 — The largest cause of neonatal death isn't a clinical cause. `death_other` is 30.7% of resolved deaths, the largest single bucket in all ten held months (28.8%–32.1%). The two named causes behind it can't be separated: birth asphyxia 23.7% vs prematurity 23.6%, a gap of 15 deaths in 9,036 — and which one leads flips between months. A bulletin ranking named clinical causes would be reporting noise. The finding is about the reporting system, not newborn medicine.

2 — A striking correlation is a confound. A composite equipment index correlates ≈ −0.87 with facility mortality, pooled across 117 facilities — reads as "equip facilities, save newborns." Stratified by tier it collapses to near zero: District −0.07, Health Centre +0.04, Provincial −0.29. The pooled number describes which tier a facility sits in, not what equipping it would do. Publishing the pooled figure alone would have pointed a Ministry's budget at a confound; the bulletin shows every covariate pooled and within tier, caveat attached, not withheld.

D1 — Problem decomposition & scoping (condensed; full document: `deliverable-1-scoping/README.md`, red-team amendment log: `decisions/0007-cross-provider-redteam-amendments.md`)

"Our data is a mess" is a symptom report, not a problem statement

Failure class	Meaning	Implies
Absence	Never captured at all	Digitisation problem
Latency	Captured, arrives too late to act on	Pipeline problem — DHIS2 runs 2–3wk behind
Distrust	Arrives on time, nobody believes it	Provenance problem — sources disagree, nobody reconciles
Last mile	Believed, never reaches a decision	Workflow problem

Week 1's job is to find which dominates. Distrust is tested first — cheapest to check, most commonly missed. One question for every interview, because the answers won't match: "*When two systems give different numbers, what happens?*"

Who I'd talk to, in what order (Week 1)

Day	Who	The question that matters
1	Solutions Manager, Country Director	Why were the use cases chosen before discovery?
1–2	**MoH Director** (sponsor)	"*Tell me about a decision last quarter you'd have made differently with better data.*" Then: "*When the numbers you're shown disagree with what you believe, which do you act on?*"
2	The bulletin analyst	Not "what do you need" — "*show me last quarter's, and open the files you built it from.*"
2–3	DHIS2/HMIS focal point	Where does the 2–3wk delay actually accrue, and how do paper facilities report today?
3	2 district health officers (1 strong, 1 weak district)	"*What did you look at immediately before your last resource decision?*"
3–4	1 hospital, 1 clinic, 1 paper-only facility — the data clerk, not the medical director	Watch month-end reporting happen
4	MoH ICT / InfoSec	Hosting, data residency, approval lead times — the classic 6-week killer

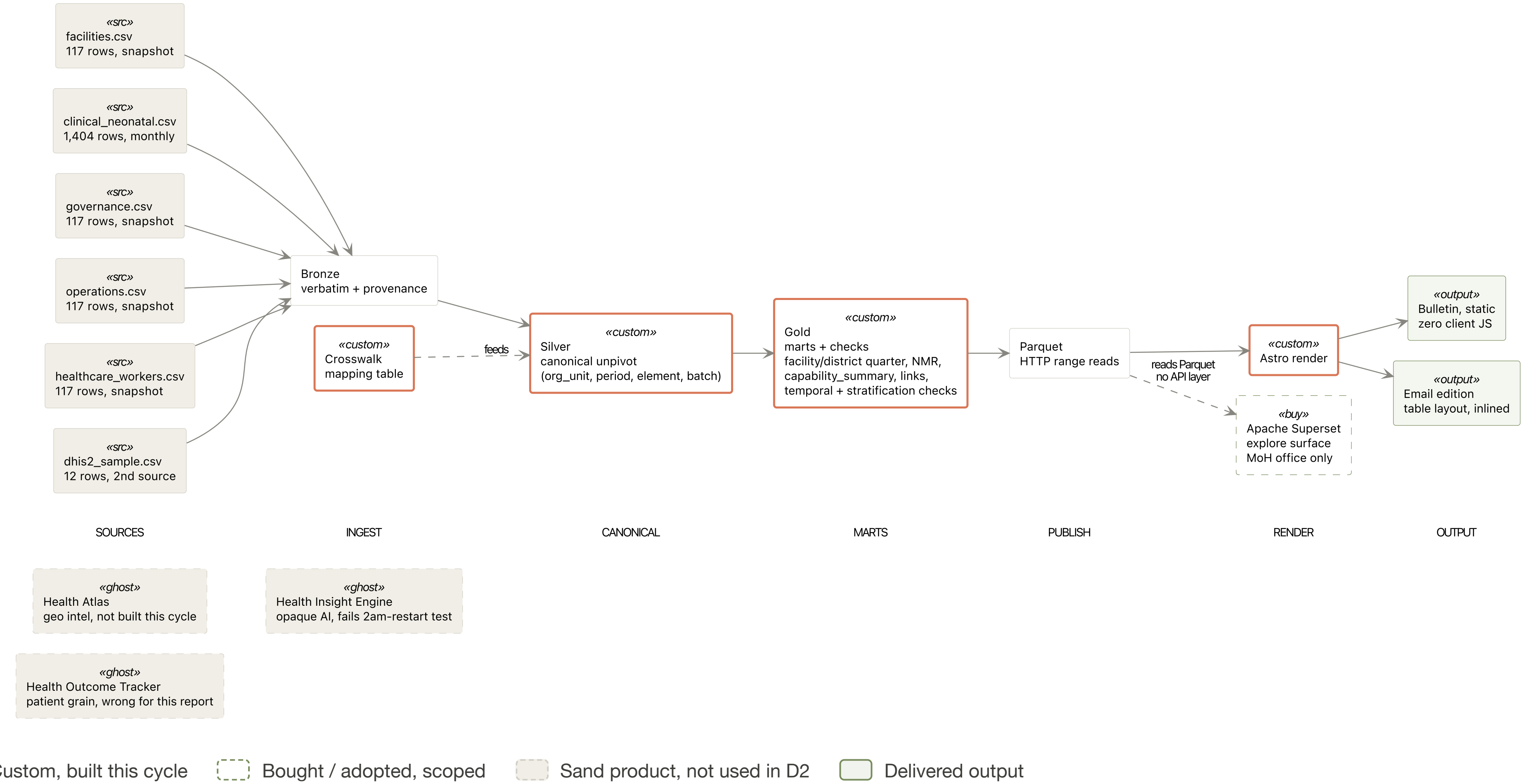
Every question is counterfactual ("what did you do last month") or observational ("show me"), never solicitational ("what would you like") — solicitation just returns a feature list.

The Week 1 gate

The commitment to A is conditional on five gates assessed at end of Week 1, not final at proposal time: **G1** Value (a decision-maker used the last bulletin for something identifiable) · **G2** Mechanisability (≥60% of cycle time is mechanical) · **G3** Inputs (DHIS2 access confirmed with a stated cutoff) · **G4** Runtime (a place to run and hand over exists) · **G5** Operator (a named receiver is assigned). G1 or G3 failing means A is the wrong commitment — said in Week 1, not Week 5.

D2 — Solution architecture: data flow, product choices, custom components

Four layers plus checks plus render (D2 §1.1, condensed). Sources → **bronze** (verbatim, stamped with lineage) → **silver** (melted, crosswalk-resolved to one canonical observation grain) → **gold** (the marts a bulletin reads, plus two seeded statistical checks that always annotate, never gate) → publish to Parquet/DuckDB → render (Astro/Vega-Lite, static bulletin + email + Superset for the connected office). One Sand product used (Superset); the crosswalk, per-figure lineage, and the two checks are custom — the rest is standard warehouse plumbing, not the reason this took six weeks.



Synthetic test data throughout (assignment CSVs plus a hand-built DHIS2 sample) — not real Rwanda health records. Mart ERD and build-vs-buy detail in the full submission and repository.

Synthetic data, built for a Sand Technologies recruitment exercise. This is not an official Ministry of Health, Rwanda publication: every figure on this page is fabricated and does not describe a real health outcome.

NEONATAL MORTALITY, 2024-Q1

50.6 per 1,000 live births

50.6 per 1,000 is 2.7× the WHO / national benchmark of 19 per 1,000.

1,819 deaths across **35,916 live births**, WHO GHO definition, 2 of 3 expected months, 2024-02 absent. 30 of 30 district figures are PROVISIONAL: 2024-01 and 2024-03 each loaded twice with differing values. Only 1 of 117 facilities stock surfactant, the standard treatment for prematurity; only 9 of 117 run an active quality-improvement programme; median self-reported reporting completeness is 76%. Capability gaps qualify whether this figure is actionable, not only whether it is settled.

Resolve the 2024-01 and 2024-03 batch conflict before treating this figure as final: DEFAULT-BATCH-01 is a placeholder default, not a decision.

MINISTRY OF HEALTH, RWANDA · NEONATAL · 2024-Q1

Quarterly Health Bulletin

2024-Q1

MONTHS HELD
2/3
2024-02 absent

FACILITIES REPORTING
117/117
in the months held

FIGURES PROVISIONAL
30/30
awaiting a batch decision

REPORTING COMPLETENESS, MEDIAN
76%
range 50% to 99%, all facilities

Provisional figures are hatched in bars, hollow in dots, underlined in tables. Every check below always renders its finding: a caveat qualifies confidence, it never withholds content. Every figure links to its lineage record.

D3 — top 5 fixes before this runs unattended, ranked by impact if skipped

1. The batch conflict has no resolution path. 2024-01 and 2024-03 each arrived twice, 234 rows, no timestamp or revision flag to arbitrate. Pipeline defaults to `DEFAULT-BATCH-01` and marks every derived figure provisional — honest, not useful, at 100% provisional. *Fix:* a triage queue where a named analyst records the choice, with who/when/why, and figures promote to settled (specified, not built). *Do not* block the pipeline on approval — a Ministry with near-zero IT capacity would then get no bulletin at all.

2. Ingestion assumes one file shape. The 71-field crosswalk was learned by hand; real DHIS2, HealthTrack, OpenMRS and paper forms won't hold still. *Fix:* LLM-proposed, human-confirmed, cached column mapping — never re-inferred per run. Free win: HXL-tagged sources map by lookup, no model call needed.

3. Chart runtime decided for the bulletin, open for the surface that doesn't exist yet. Flint → Vega-Lite adopted and working; it replaced Observable Plot outright. *Do not* default the unbuilt explore surface to whichever runtime is closest to hand when that work starts — decide explicitly, in writing, first.

4. Nothing runs on a schedule, no one is told when it fails. Hamilton already exposes the DAG as a callable — orchestration is a wrapper, not a rewrite. What matters more than the scheduler: publication must gate on the 5 checks passing, and a failed run must reach a person.

5. Facility geography is unusable. GPS is uniform-random inside Rwanda's bounding box for all 117 facilities; the bulletin honestly maps at district grain. *Fix:* join HDX's Rwanda Healthsites layer (1,345 real facilities, open licence). *Do not* reverse-geocode district centroids as a substitute — that manufactures false precision.

Already hardened, not just flagged: disclosed-not-withheld statistical caveats (temporal-signal and stratification checks always annotate, never gate); cross-surface figure agreement (`check-agreement.mjs`, 7 shared figures); filename/content mismatch refused at publish; chart-palette drift against the stylesheet (`check-tokens.mjs`); per-figure lineage recorded for every published number.

D4 — Handover

The deliverable is not the artifact, it's a named Digital Health Officer independently restarting the pipeline once, unassisted, before exit. Two training sessions: Session 1 (60–90min) the DHO drives the full cycle hands-on-keyboard against last quarter's real data, reading every check pass rather than trusting a green tick; Session 2 (45–60min) diagnoses a deliberately seeded failure — a renamed CSV column, a check made to fail on purpose — using only the runbook's recovery table. Success is naming which document to open, not fixing that specific break.

Full clean-rebuild verified 2026-08-11 from a wiped state (`mart`, `web/dist`, `node_modules`, `.venv` all removed): sync → rebuild (10 parquet files) → publish (both quarters) → verify — 5/5 checks passed, output reproduced at the same byte hash. That run predates later pipeline changes (ADR-0012) and has **not been re-verified against current code** — flagged here, not hidden; re-running and diff-checking the hash is the next action before reproducibility can be re-claimed.

Three documents (`deliverable-4-handover/runbook.md` · `data-contract.md` · `decision-register.md`) answer "what would you hand the MoH IT team" and ship in the repository, not this PDF. **Full detail for all four deliverables, the pipeline code, and the openspec behaviour specs: github.com/nyashkn/sand-fde-int**